

SmolVLM for Dense Video Captioning

Week 5,6: Internship Update

TvSum and SumMe

- TvSum(<https://ieeexplore.ieee.org/document/7299154>):
 - 50 user-generated videos of the length 2-10 minutes collected from YouTube, spanning 10 diverse categories such as changing a vehicle tire and grooming an animal
 - Each video is automatically segmented into non-overlapping shots using visual content change
 - Twenty human annotators rated the importance of each shot on a scale from 1 to 5, reflecting how essential each shot is for summarizing the video
 - The final importance score of a shot is the average across annotators
 - For frame-level supervision, the same importance score is assigned to all frames within a shot

TvSum and SumMe

- SumMe(<https://paperswithcode.com/dataset/summe>):
 - 25 videos of the length 1-6 minutes annotated by at least 15 people
 - Consists of videos with different contents (e.g., holidays, events, sports) and various types of camera angles (e.g., static, egocentric, or moving cameras)
 - the ground truth are importance scores $[0,1]$

Video Summarization Task

- SumMe:
<https://paperswithcode.com/sota/supervised-video-summarization-on-summe?p=csta-cnn-based-spatiotemporal-attention-for>
- TvSum:
<https://paperswithcode.com/sota/supervised-video-summarization-on-tvsum?p=csta-cnn-based-spatiotemporal-attention-for>
- Metrics:
 - F1 score
 - Sperman's coefficient of correlation
 - Kendall's coefficient of correlation (captures monotonic relationship)

Supervised Video Summarization via Multiple Feature Sets with Parallel Attention

(<https://paperswithcode.com/paper/supervised-video-summarization-via-multiple>)

- MSVA (Multi-Source Visual Attention)
- Trained on SumMe and TvSum
- Ranked 7th on video summarization on SumMe
- New cross-validation approach

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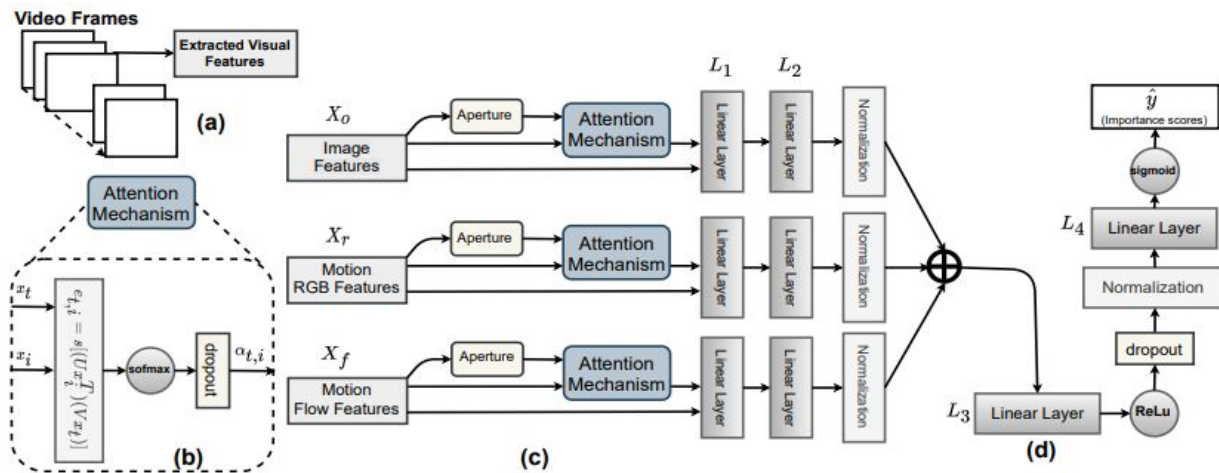


Fig. 1. The neural network architecture for the Multi-Source Visual Attention (MSVA) model with parallel self-attention mechanism based on multiple feature sets

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- Content-based Image and Motion Features

- Three visual features
- 2 frames per second
- Image content (X_0): GoogleNet trained on ImageNet; features extracted from pool5 layer (dimension: 1024)
- Motion: pretrained I3D (Inflated 3D ConvNet) trained on Kinetics dataset:
 - RGB features (X_r): channel-wise color information with regard to scene changes
 - Optical flow (X_f): optical flow
 - extracted from second to last layer (dimension: 1024)

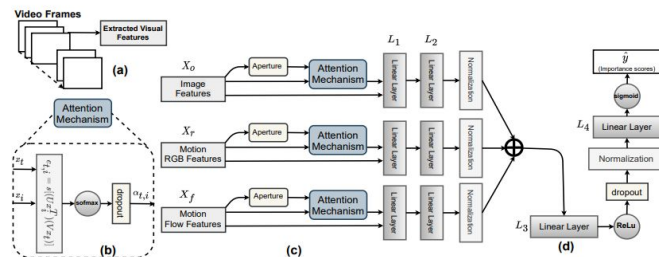


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● Attention Module

- Separate attention block for each feature
- 2 linear layers and a single normalization block
- Aperture: $[t-p, t+p]$ (window)
- The attention weights at the time index t with a subset from the feature set X are:
- Normalized attention vectors.

$$e_{t,i} = \frac{\exp(e_{t,i})}{\sum_{k=0}^{T-1} \exp(e_{t,k})}$$

- $\alpha_t = (\alpha_{t,t-p}, \dots, \alpha_{t,t}, \dots, \alpha_{t,t+p}) \quad Z = \text{Norm}(L_2(L_1(\alpha_t X_t)))$

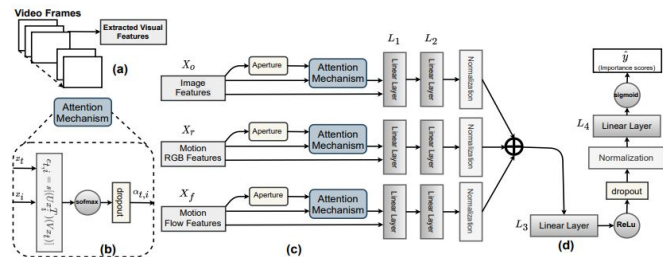


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- Multi-source fusion: intermediate fusion, other tested

$$h = L_4(Norm(Drop(ReLU(L_3(Z_o + Z_f + Z_r))))))$$

$$\hat{y} = sigmoid(h)$$

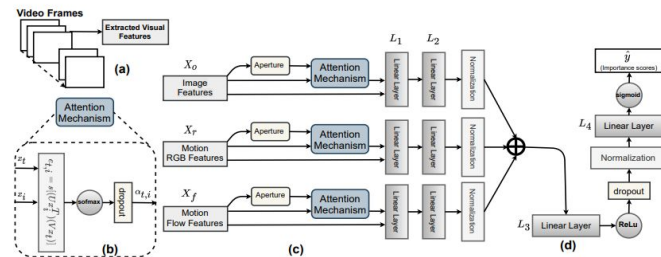


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- The evaluation has previously been done based on 5-fold cross-validation and the reported results are F1 scores averaged across the five splits of the corresponding datasets
- Some videos are excluded, some are repeated
- New 5 non-overlapping splits
- F1 score, Spearman's coefficient, Kendall's coefficient

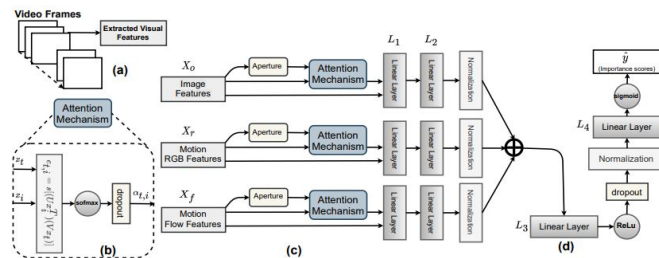


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CSTA: CNN-based Spatiotemporal Attention for Video Summarization (<https://arxiv.org/pdf/2405.11905>)

- Spatial and temporal attention
- On SumMe (Kendall's tau metric)
State of the Art for video summarization
- On TvSum (Kendall's tau metric)
second for video summarization
- Needs fewer multiply-accumulate operations

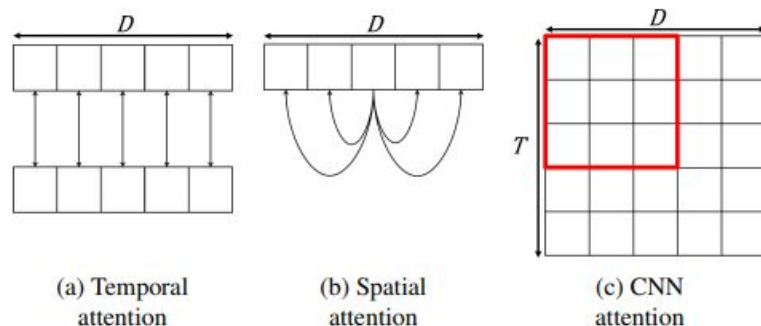


Figure 1. Approaches for calculating attention. Each row is the feature vector of a frame. T is the number of frames, and D is the dimension of the feature.

CSTA: CNN-based Spatiotemporal Attention for Video Summarization (<https://arxiv.org/pdf/2405.11905>)

- Trained on SumMe and TVSum-the output is an importance score of a frame
- 9,779,851parameters (plus embedding ~6.8 million)

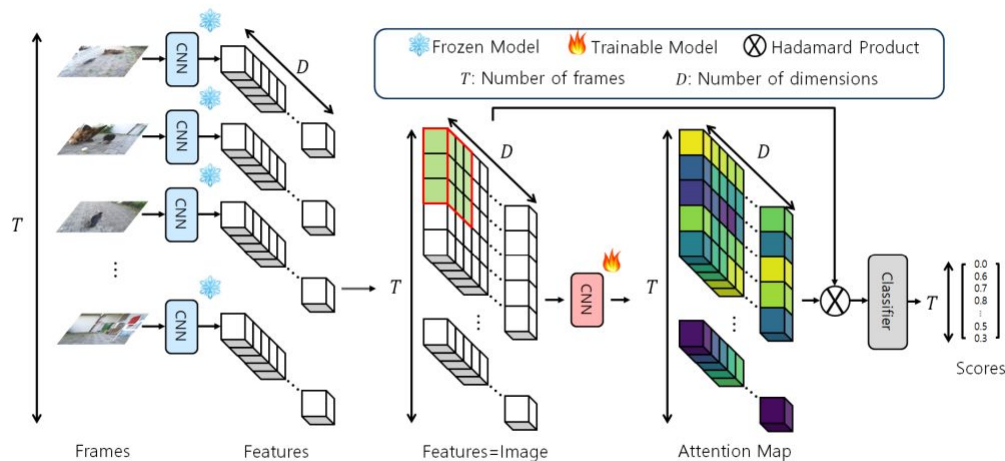


Figure 2. Workflow of CSTA

CSTA: CNN-based Spatiotemporal Attention for Video Summarization (<https://arxiv.org/pdf/2405.11905>)

- Embedding Process:
 - frozen pre-trained CNN model (GoogleNet or some other) modifies the whole dataset to the size $R^{\wedge}(T \times D)$
 - Concatenated classification token

$$X'' = \text{Concat}_{axis=0}(X', X', X')$$

$$E = \text{Concat}_{axis=1}(X_{CLS}, X'')$$

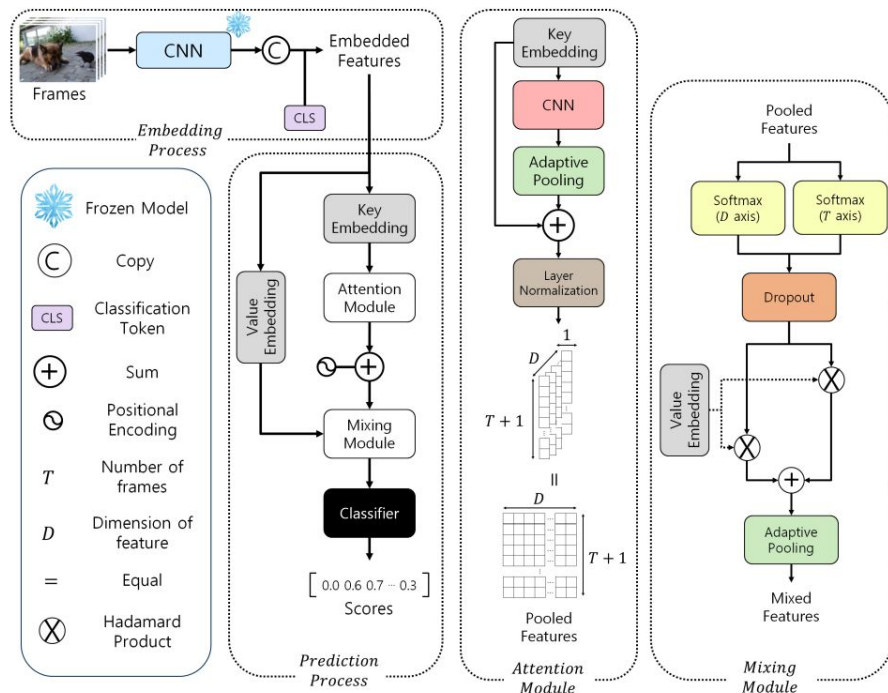


Figure 3. Architecture of CSTA

CSTA: CNN-based Spatiotemporal Attention for Video Summarization (<https://arxiv.org/pdf/2405.11905>)

- Prediction Process:
 - Attention Module + Mixing Module
 - Attention Module makes attention maps from E and Mixing Module aggregates this attention with E

Algorithm 1: Prediction Process

input : $E \in \mathbb{R}^{3 \times (T+1) \times D}$
output: $S \in \mathbb{R}^T$

```

1  begin
2     $E^K = W^K E$ 
3     $E^V = W^V E[0]$ 
4
5     $P = \text{Attention Module}(E^K)$       Section 3.4
6     $P_{pos} = P + \text{Positional Encoding}$ 
7     $M = \text{Mixing Module}(P_{pos}, E^V)$   Section 3.5
8     $S = \text{Classifier}(M)$ 
9  return S
10 end
    
```

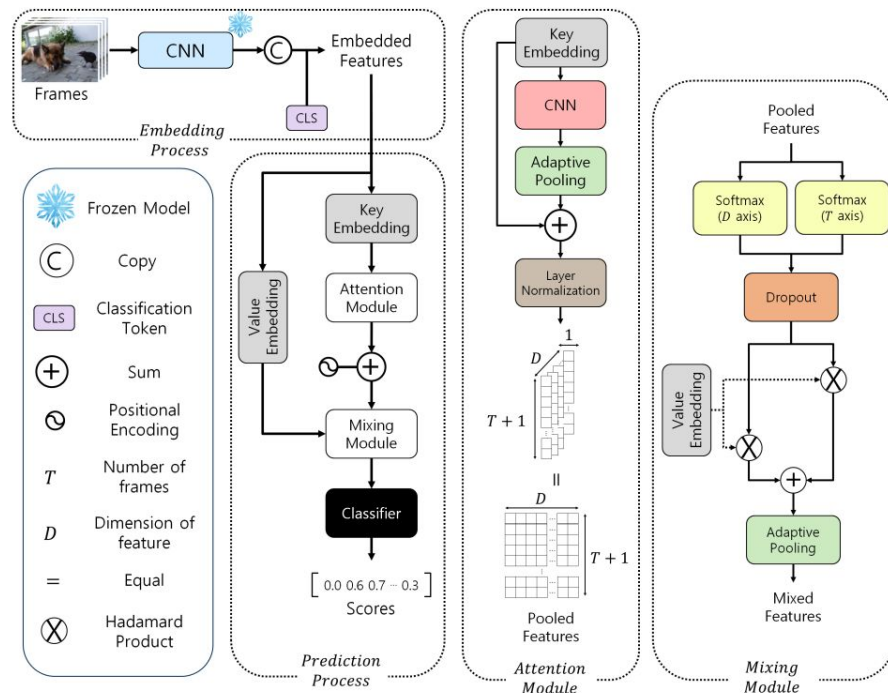


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CSTA: CNN-based Spatiotemporal Attention for Video Summarization (<https://arxiv.org/pdf/2405.11905>)

- Attention Module
 - Attention Module produces attention maps by utilizing a trainable CNN (GoogleNet)
 - Adaptive Pooling: bilinear interpolation

$$P = \text{LayerNorm}(E_{\text{pool}}^K + E^K[0])$$

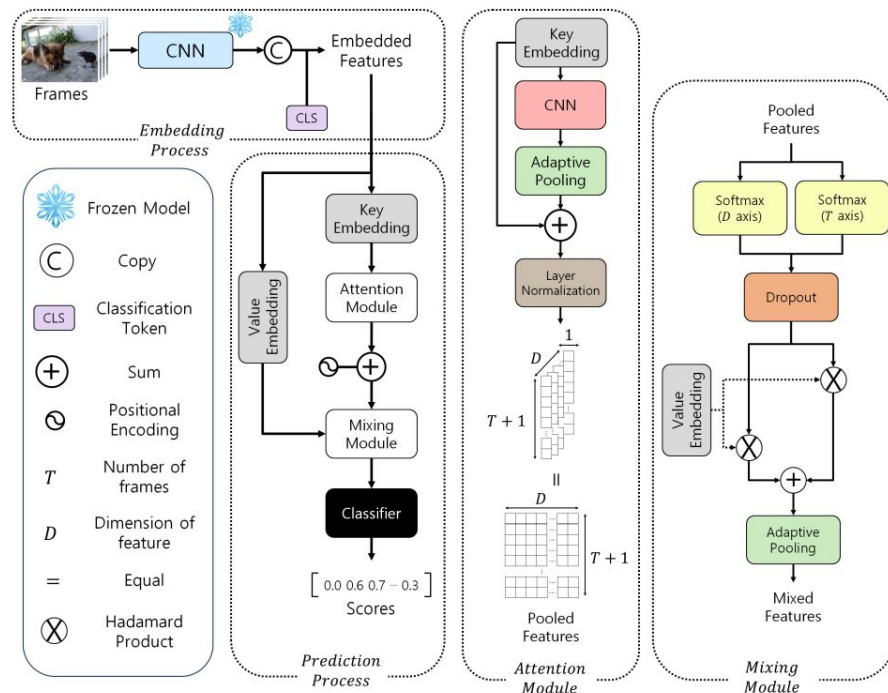


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CSTA: CNN-based Spatiotemporal Attention for Video Summarization (<https://arxiv.org/pdf/2405.11905>)

● Mixing Module

- Softmax along the time and dimension axes to compute the temporal and visual weighted values

$$Att_T : \sigma(d_i) = \left(\frac{e^{d_i}}{\sum_j e^{d_j}} \right) j = 1, \dots, T + 1$$

$$Att_D : \sigma(d_k) = \left(\frac{e^{d_k}}{\sum_l e^{d_l}} \right) k = 1, \dots, D$$

- Dropout
- Element-wise multiplication

$$M = Att_T \odot E^V + Att_D \odot E^V$$

- Adaptive pooling (average): back to $R^{\wedge}(T \times D)$

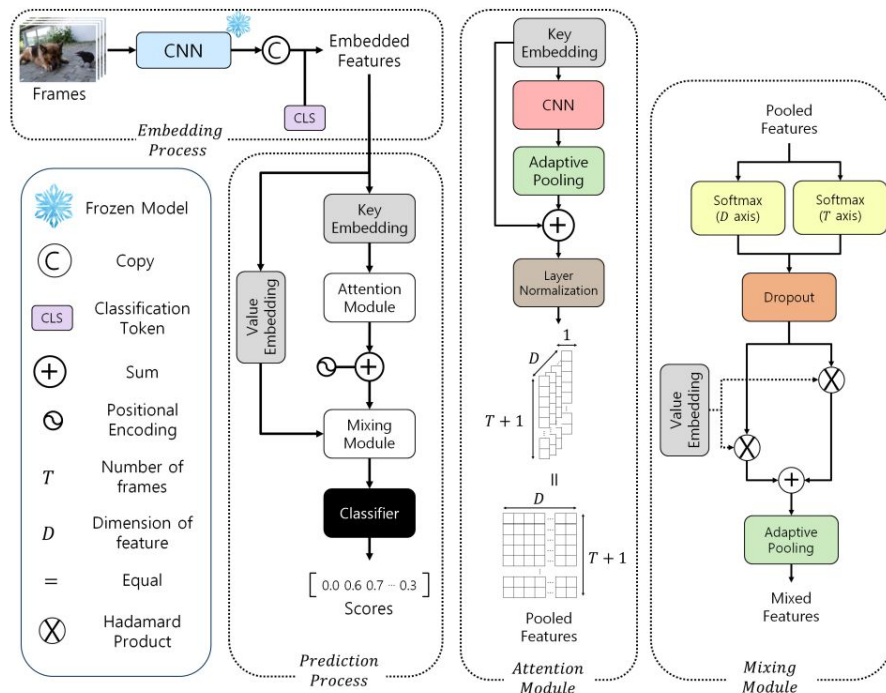


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CSTA: CNN-based Spatiotemporal Attention for Video Summarization (<https://arxiv.org/pdf/2405.11905>)

- Classifier:

$$R = \text{LayerNorm}(\text{Dropout}(\text{ReLU}(\text{FC}(M'))))$$

$$S = \text{Sigmoid}(\text{FC}(R))$$

- every 15th frame is taken
- KTS derives shots and assigns them average importance scores
- the most 'important' shots are extracted
- knapsack algorithm:

$$\max \sum S_i$$

$$\sum \text{Length}_i \leq 15\%$$

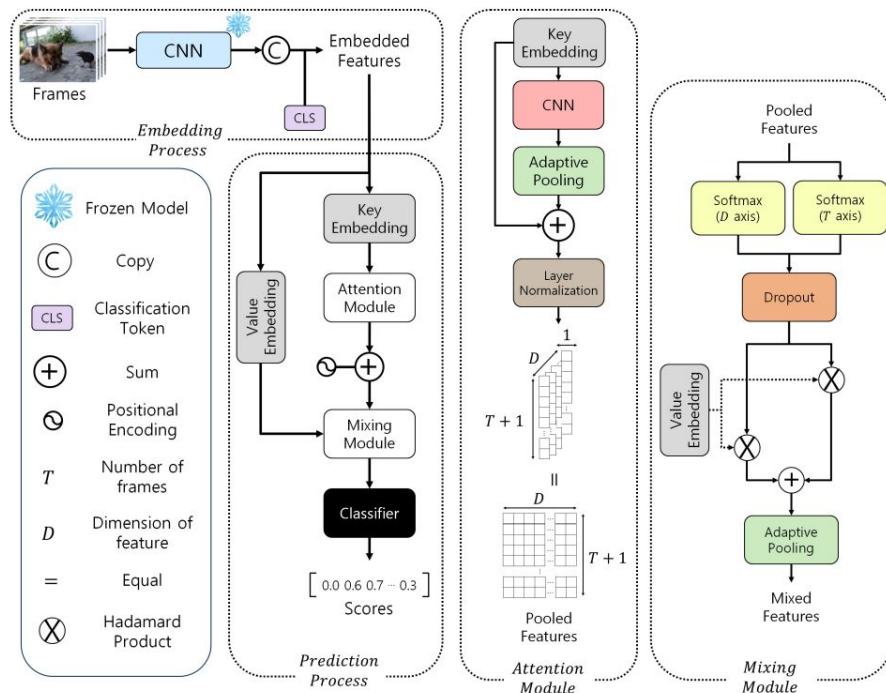


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CSTA: CNN-based Spatiotemporal Attention for Video Summarization (<https://arxiv.org/pdf/2405.11905>)

- Evaluation metrics: Kendall's and Spearman's coefficients: 'Due to the limitation of the summary length, the F1 score is evaluated to be higher if models choose as many short shots as possible and ignore long key shots.'
- For attention (and CNN embeddings) they use: MobileNet-V2, EfficientNet-B0, GoogleNet, and ResNet-18

Category-Specific Video Summarization

(https://link.springer.com/chapter/10.1007/978-3-319-10599-4_35)

- Kernel Temporal Segmentation (KTS)
 - non-abrupt boundaries between consecutive segments with different semantic content
 - used for noisy stepwise constant signal in order to track steps, considers a whole signal at once

$$K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R},$$

$$K(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle_{\mathcal{H}},$$

$$J_{m,n} = L_{m,n} + C \cdot g(m, n),$$

$$v_{t_i, t_{i+1}} = \sum_{t=t_i}^{t_{i+1}-1} \|\phi(x_t) - \mu_i\|_{\mathcal{H}}^2,$$

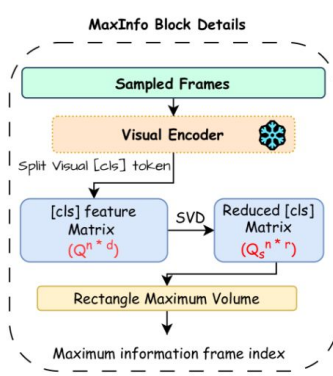
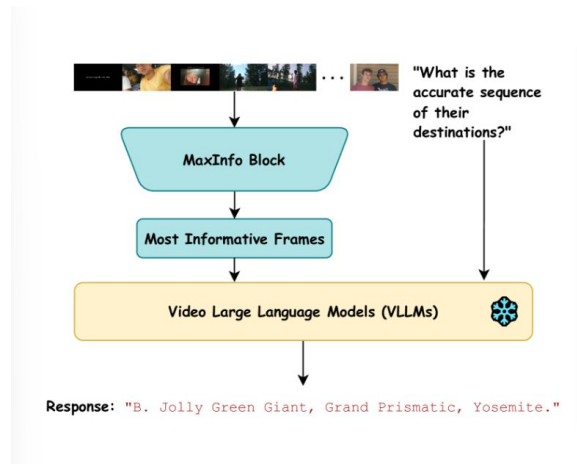
$$\mu_i = \frac{1}{t_{i+1} - t_i} \sum_{t=t_i}^{t_{i+1}-1} \phi(x_t).$$

$$L_{m,n} = \sum_{i=0}^{m-1} v_{t_i, t_{i+1}}.$$

$$g(m, n) = m \left(\log \left(\frac{n}{m} \right) + 1 \right).$$

MaxInfo: A Training-Free Key-Frame Selection Method Using Maximum Volume for Enhanced Video Understanding (<https://arxiv.org/html/2502.03183v2>)

- Code implemented, a need for the number of frames



Algorithm 1 MaxInfo Block: SVD + MaxVol for Keyframe Selection

- 1: **Input:** A set of n frames $\mathbf{I} = \{I_1, I_2, \dots, I_n\}$
- 2: **Embedding:** Convert each frame I_j into a [CLS] embedding:
$$q_n = \text{flatten}(\text{clip_model}(I_n)), Q = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix} \in \mathbb{R}^{n \times d}.$$
- 3: **SVD Reduction:** Perform truncated SVD on Q .
$$Q \approx U_r \Sigma_r V_r^T \rightarrow Q_s = U_r \in \mathbb{R}^{n \times r}.$$
- 4: **MaxVol Selection:** Run $\text{rect_maxvol}(Q_s, \text{tol})$ to find pivot indices:
$$\text{piv} = \text{rect_maxvol}(Q_s, \text{tol}),$$
identifying rows (frames) that span the reduced embedding space.
- 5: **Output:** Indices **piv** of the most informative keyframes.